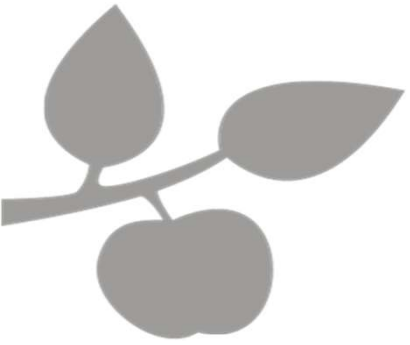




DM583

Data Mining

Hierarchical Clustering

Why Hierarchical Clustering ?

- Oftentimes, we don't want to look at a single partition of the dataset with k clusters, but rather explore a whole spectrum of partitions at different levels of granularity, corresponding to different numbers of clusters
- In other words, we may want a **clustering hierarchy**, rather than a single partition
- The reasons are manyfold:
 - By building a hierarchy one doesn't need to specify the number of clusters k in advance
 - Rather, a hierarchical structure may help determine the best k (if any) a posteriori
 - The hierarchical structure may also reveal that the data is naturally organised hierarchically
 - sub-clusters inside clusters
 - The hierarchy provides a powerful **visualisation** tool of high-dimensional data
 - which may reveal not only clusters and sub-clusters, but also potential **outliers**

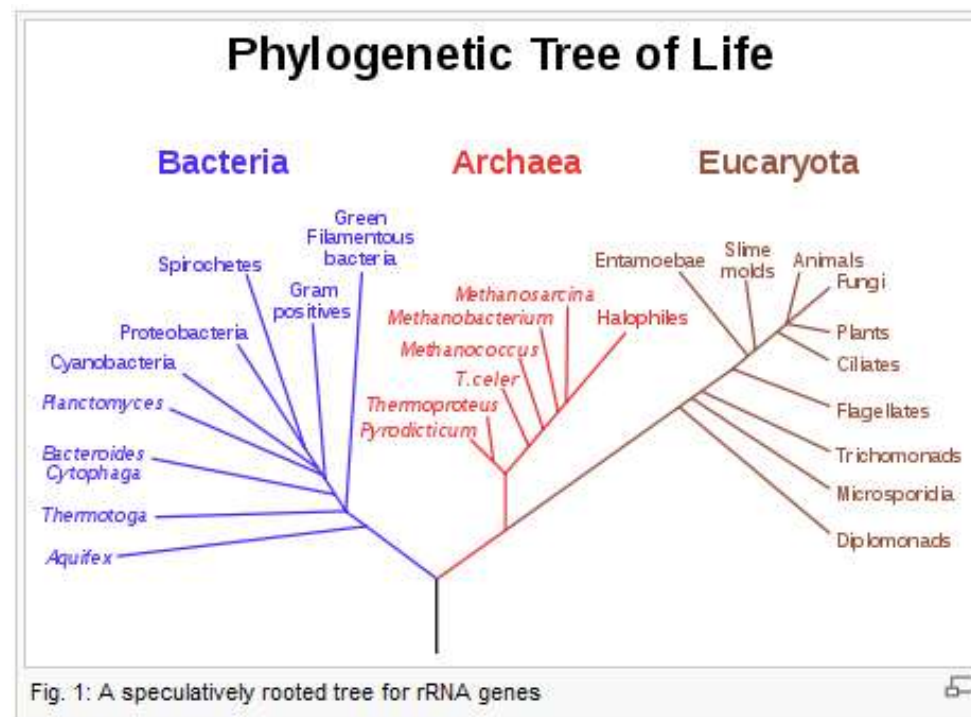
Clustering Hierarchies

- **Hierarchy** (of data partitions):
 - **Sequence of nested hard partitions**
 - A partition P_1 is *nested* within P_2 if each component (cluster) of P_1 is a subset of a component of P_2
 - **Example:**
$$P_1 = \{ (x_1), (x_3, x_4, x_6), (x_2, x_5) \}$$
$$P_2 = \{ (x_1, x_3, x_4, x_6), (x_2, x_5) \}$$
 - **Counter-Example:**
$$P_3 = \{ (x_1, x_3, x_4, x_6), (x_2, x_5) \}$$
$$P_4 = \{ (x_1, x_2), (x_3, x_4, x_6), (x_5) \}$$

Clustering Hierarchies

- A complete hierarchy:
 - Starts or ends with a completely *disjoint partition/clustering*
 - **Disjoint clustering**: contains only **atomic clusters (singletons)**
 - Example: $\mathbf{P} = \{ (\mathbf{x}_1), (\mathbf{x}_2), (\mathbf{x}_3), (\mathbf{x}_4), (\mathbf{x}_5), (\mathbf{x}_6) \}$
 - It is also called “trivial clustering solution”
 - Starts or ends with a single (not partitioned) component
 - Dataset itself as a single “cluster” with all data objects
 - Example: $\mathbf{P} = \{ (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6) \}$
 - Generally, there are $N - 2$ intermediate partitions

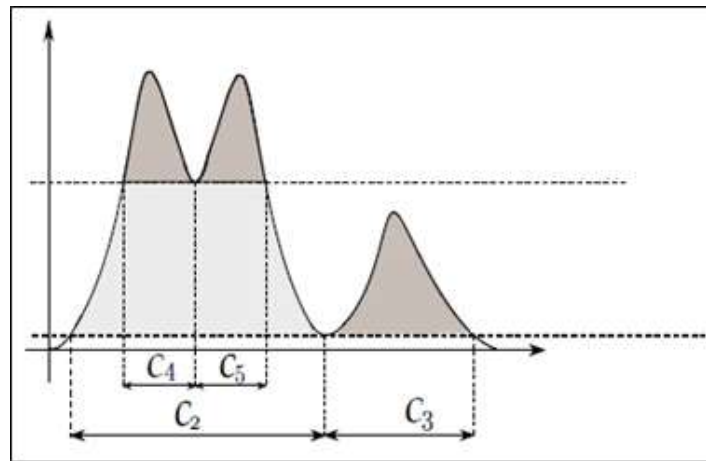
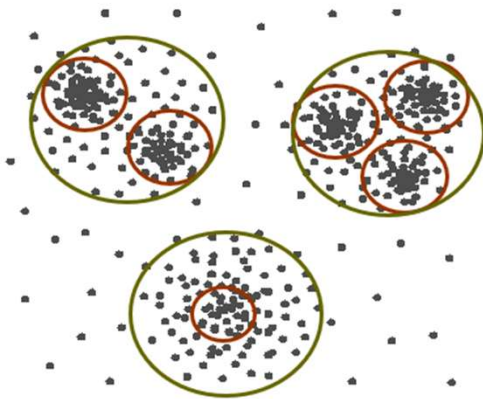
- **Hierarchies** are commonly used to organise information:
 - Categories and subcategories
 - Example: Phylogenetic Trees in Biology



http://en.wikipedia.org/wiki/Phylogenetic_tree

- The relationship between natural clusters and subclusters in many datasets is intrinsically **hierarchical**:

- Examples:



Hierarchical Clustering as a Relational Approach

Hierarchical Clustering can
operate with (dis)similarities
only: they are (or can be turned
into) **relational algorithms**

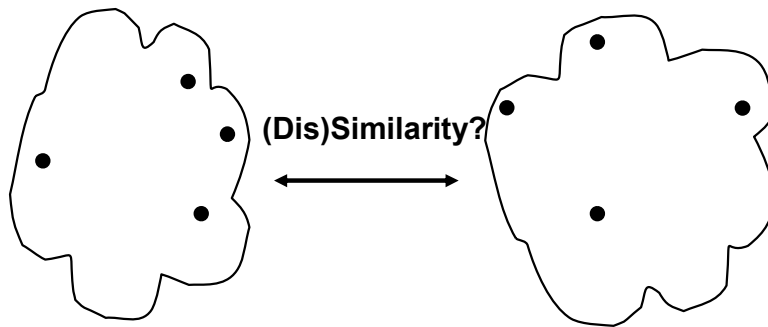
0	8	8	7	7
	0	2	4	4
		0	3	3
			0	1
				0

Classic Agglomerative Hierarchical Clustering Algorithm (AHC)

1. Start with each observation being a cluster on its own (i.e., a **singleton**), and compute all pairwise distances between observations (if not given as input)
2. Find the **closest pair** of current clusters and merge them into a single cluster
3. Compute the **distance** between this newly born cluster and the other clusters
4. Repeat Steps 2 and 3 until a single cluster remains

There is a number of algorithms that follow exactly the same steps above. The difference between them is how the **distances between clusters** are computed

How to Define Inter-Cluster (Dis)Similarity

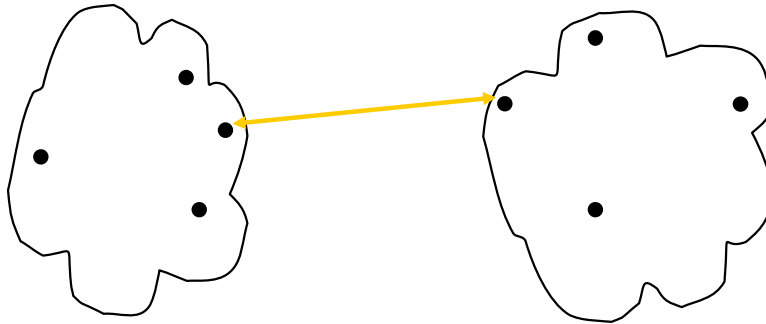


- MIN
- MAX
- Average
- Distance Between Centroids
- Other methods
 - Ward's
 - ...

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						

· **Proximity Matrix**

How to Define Inter-Cluster (Dis)Similarity

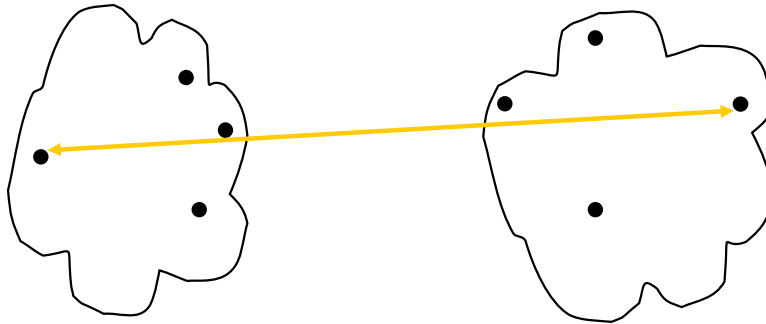


- MIN
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	p1	p2	p3	p4	p5	...
p1						
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·
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· **Proximity Matrix**

How to Define Inter-Cluster (Dis)Similarity

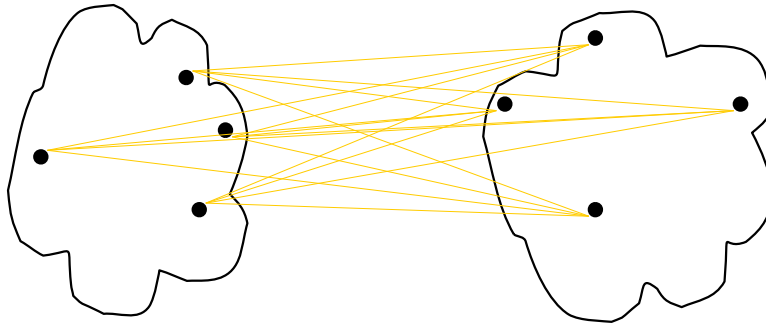


- MIN
- **MAX**
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	p1	p2	p3	p4	p5	...
p1						
p2						
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.						

· **Proximity Matrix**

How to Define Inter-Cluster (Dis)Similarity

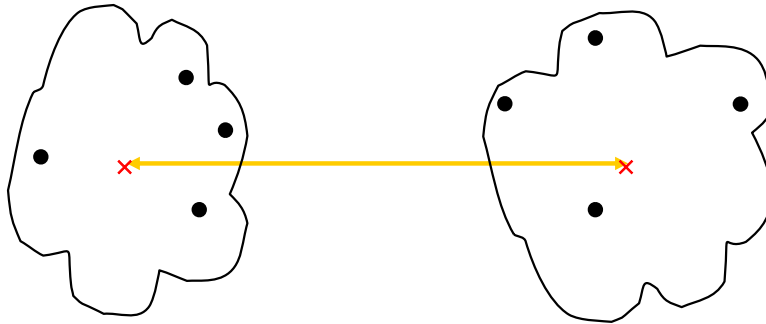


- MIN
- MAX
- **Average**
- Distance Between Centroids
- Other methods
 - Ward's
 - ...

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
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Proximity Matrix

How to Define Inter-Cluster (Dis)Similarity



- MIN
- MAX
- Average
- Distance Between Centroids
- Other methods
 - Ward's
 - ...

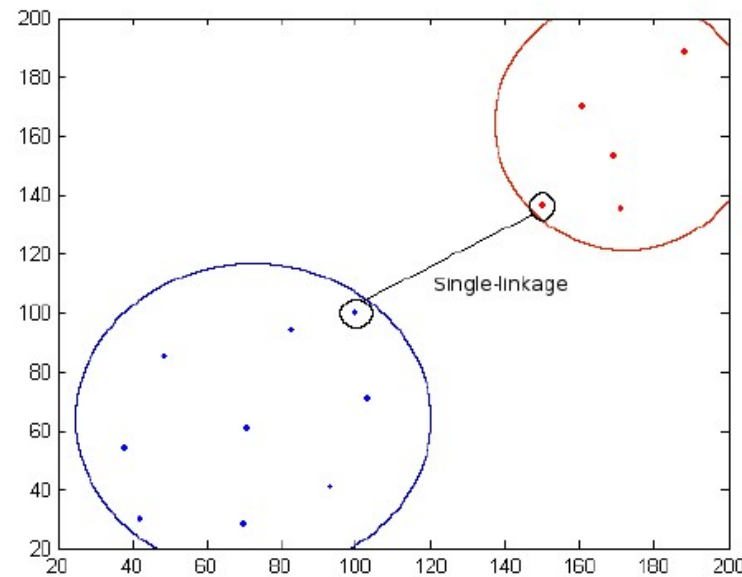
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
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Proximity Matrix

Agglomerative Hierarchical Clustering Algorithm

- **Single-Linkage:**

- Distance between two clusters is measured as the smallest distance between any two observations



Ricardo Campello

Computation of Single-Linkage

- Useful property of min:
 - $\min\{\mathbf{D}\} = \min\{ \min\{\mathbf{D}_1\} , \min\{\mathbf{D}_2\} \}$
 - $\mathbf{D}$, $\mathbf{D}_1$ and $\mathbf{D}_2$ are real-valued sets such that $\mathbf{D}_1 \cup \mathbf{D}_2 = \mathbf{D}$
 - Example:
 - $\min\{10, -3, 0, 100\} = \min \{ \min\{10, -3\}, \min\{0, 100\} \} = -3$
 - Property holds recursively (for $\min\{\mathbf{D}_1\}$ and $\min\{\mathbf{D}_2\}$)
- Why can this property be useful for Single-Linkage ?
 - Given the distances between a cluster **A** and two clusters **B** and **C** that have been merged
 - It is trivial to compute the distance between **A** and $(\mathbf{B} \cup \mathbf{C})$ from the previous distances

Single-Linkage (Example):

- Initial distance matrix ($\mathbf{D}_1$) for 5 observations {1, 2, 3, 4, 5}: first merge takes place between singletons 1 and 2 (closest clusters)

$$\mathbf{D}_1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & & & & \\ & 2 & & & \\ & 6 & 5 & & \\ & 10 & 9 & 4 & 0 \\ & 9 & 8 & 5 & 3 & 0 \end{bmatrix} \end{matrix}$$

- Updating the distances (between the new cluster and the others):

$$d_{(12)3} = \min\{d_{13}, d_{23}\} = d_{23} = 5;$$

$$d_{(12)4} = \min\{d_{14}, d_{24}\} = d_{24} = 9;$$

$$d_{(12)5} = \min\{d_{15}, d_{25}\} = d_{25} = 8;$$

- The result is a new distance matrix ($\mathbf{D}_2$)

$$\mathbf{D}_2 = \begin{matrix} & 12 \\ \begin{matrix} 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & & & \\ 5 & 0 & & \\ 9 & 4 & 0 & \\ 8 & 5 & \boxed{3} & 0 \end{bmatrix} \end{matrix}$$

- The closest clusters now are (singletons) 4 and 5
- Merging **4** and **5** yields three clusters: {1,2}, {4,5}, {3}
- Since $d_{(12)3}$ is already available, we can update the distances as:

$$d_{(45)(12)} = \min\{d_{4(12)}, d_{5(12)}\} = d_{5(12)} = 8$$

$$d_{(45)3} = \min\{d_{43}, d_{53}\} = d_{43} = 4$$

which yields the following new matrix:

$$\mathbf{D}_3 = \begin{matrix} & 12 \\ \begin{matrix} 3 \\ 45 \end{matrix} & \begin{bmatrix} 0 & & & \\ 5 & 0 & & \\ 8 & \boxed{4} & 0 & \end{bmatrix} \end{matrix}$$

Merge *clusters* {3} and {4,5};
Finally, merge {3,4,5} and {1,2} into a single cluster

■ **Note on Computational Speed-Up of Single Linkage:**

The (dis)similarity between 2 clusters follows immediately from the (dis)similarity matrix updated in the previous iteration, such that *there is no need to resort to the original matrix*

- For instance, in the previous example we simplified the computation of $d_{(12)(45)}$ as $\min\{d_{(12)(4)}, d_{(12)(5)}\}$ by making use of the min property:

- $\min\{d_{(12)(4)}, d_{(12)(5)}\} = \min\{9, 8\} = \min\{d_{14}, d_{24}, d_{15}, d_{25}\}$

- The sequence of nested partitions obtained in the previous example was:

$\{ (1), (2), (3), (4), (5) \} \rightarrow \{ (1, 2), (3), (4), (5) \} \rightarrow$

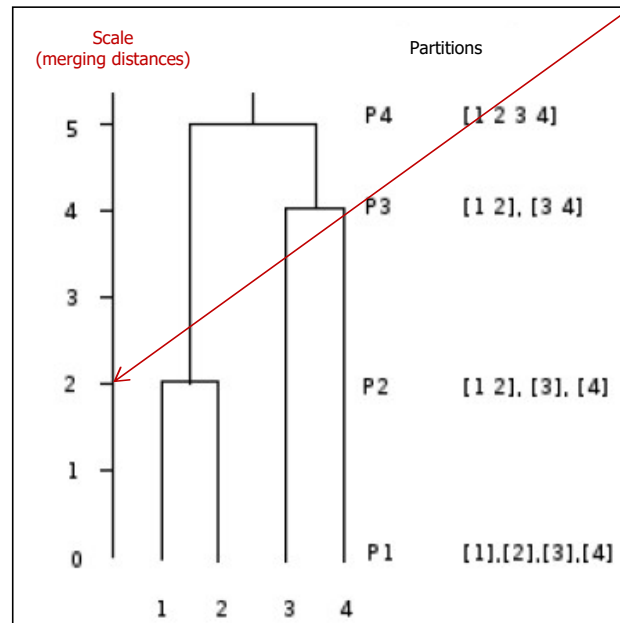
$\{ (1, 2), (3), (4, 5) \} \rightarrow \{ (1, 2), (3, 4, 5) \} \rightarrow \{ (1, 2, 3, 4, 5) \}$

- With this collection of nested partitions, as well as the distances corresponding to the pairs of merged clusters, we can build a **dendrogram**
 - Dendrogram = visual representation of the **clustering hierarchy** enhanced with a **scale** of critical distances between clusters
 - A powerful tool for data *visualisation* and *exploratory data analysis*

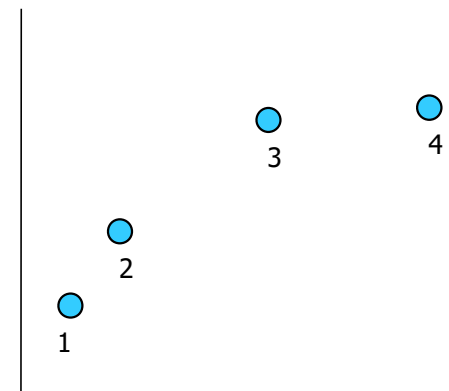
Dendrogram (Example):

$$\mathbf{D} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 2 & 7 & 13 \\ 2 & 0 & 5 & 10 \\ 7 & 5 & 0 & 4 \\ 13 & 10 & 4 & 0 \end{bmatrix} \end{matrix}$$

merging distances
(between **clusters**)



Dendrogram



dataset

Exercise

- Draw the complete **dendrogram** (hierarchy + vertical axis/scale) for one of our previous examples of Single-Linkage:

$$\mathbf{D}_1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & & & & \\ 2 & 0 & & & \\ 6 & 5 & 0 & & \\ 10 & 9 & 4 & 0 & \\ 9 & 8 & 5 & 3 & 0 \end{bmatrix} \end{matrix}$$

Dendrograms and Partitions

Partitions can be obtained by "cutting through" the dendrogram

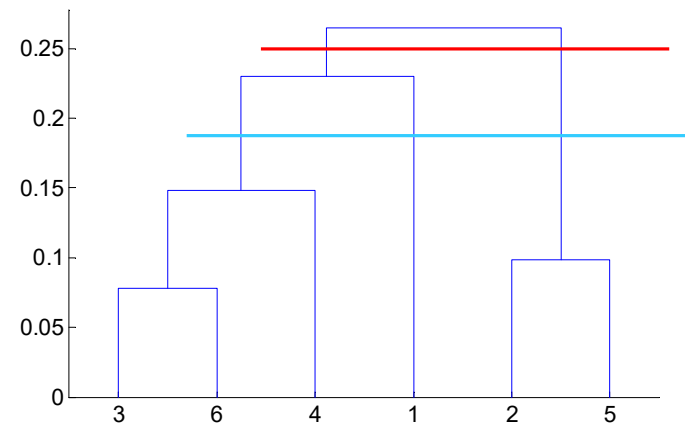
Horizontal cuts

no. of clusters = no. of line intersections

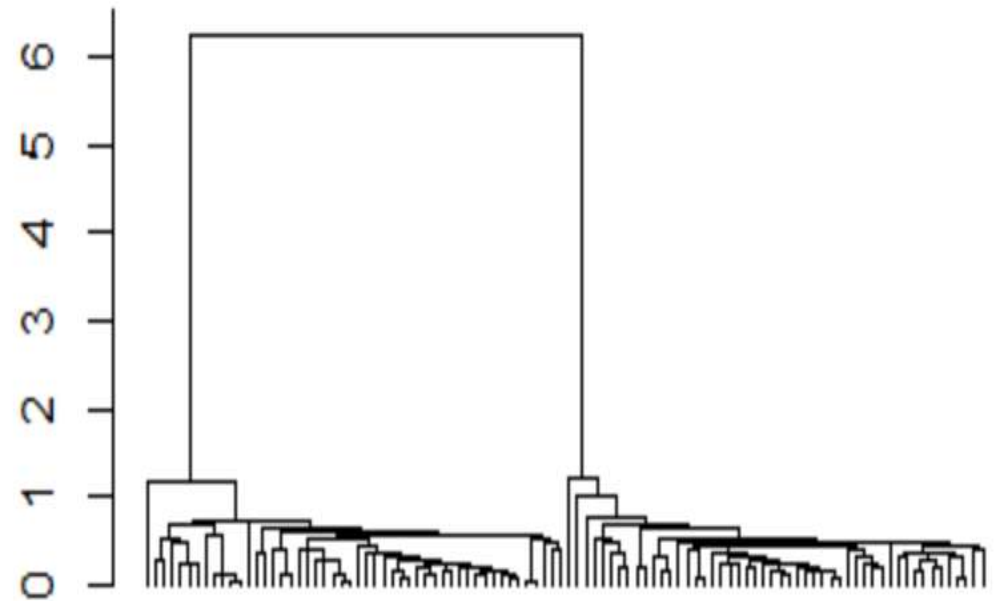
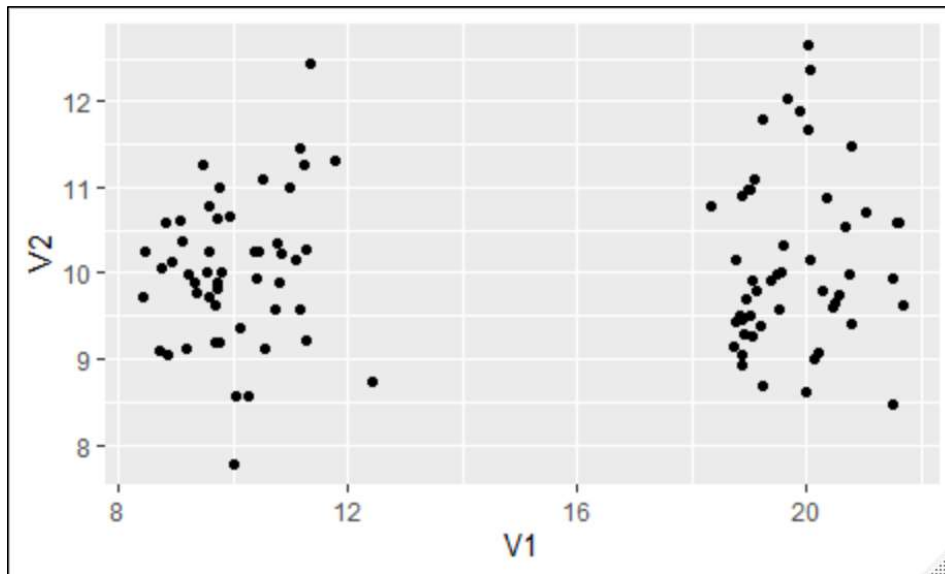
Examples:

$$P_2 = \{ (\mathbf{x}_1, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_6), (\mathbf{x}_2, \mathbf{x}_5) \}$$

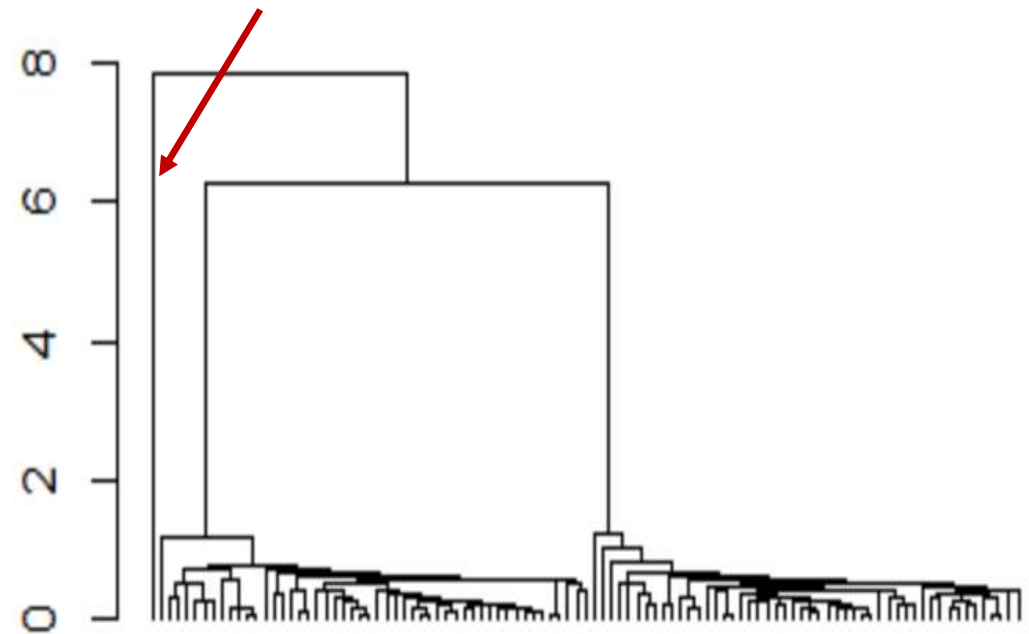
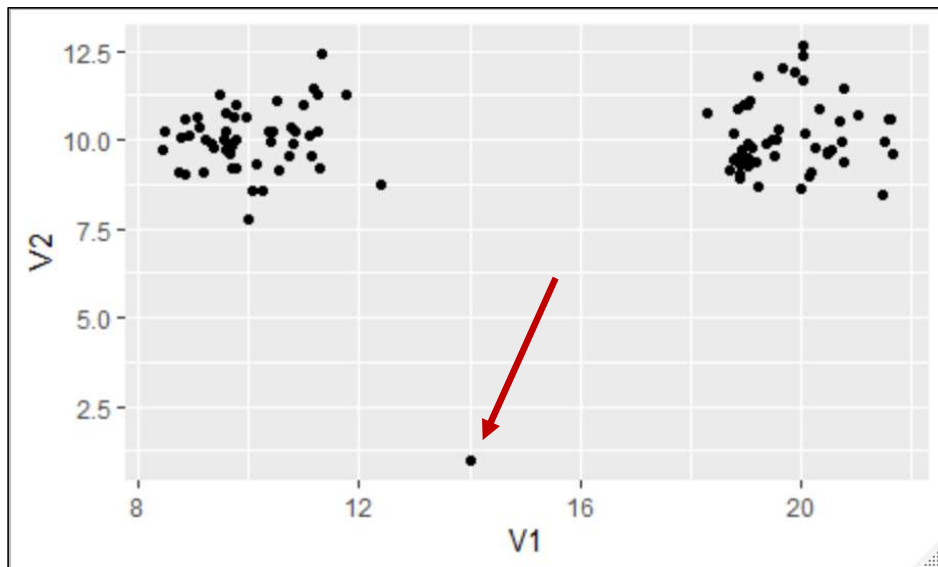
$$P_1 = \{ (\mathbf{x}_1), (\mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_6), (\mathbf{x}_2, \mathbf{x}_5) \}$$



A dendrogram may visually indicate the number of natural clusters:



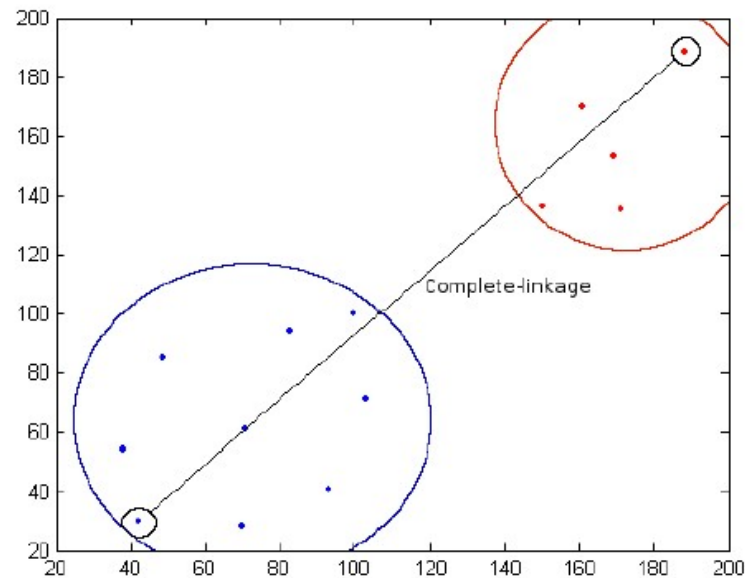
A dendrogram may also indicate the presence of outliers:



Agglomerative Hierarchical Clustering Algorithm

- **Complete-Linkage (Max):**

- Distance between two clusters is measured as the largest distance between any two observations



Ricardo Campello

Computation of Complete-Linkage

- Useful property of max:

- $\max\{\mathbf{D}\} = \max\{ \max\{\mathbf{D}_1\} , \max\{\mathbf{D}_2\} \}$

- $\mathbf{D}$, $\mathbf{D}_1$ and $\mathbf{D}_2$ are real-valued sets such that $\mathbf{D}_1 \cup \mathbf{D}_2 = \mathbf{D}$

- Example:

- $\max\{10, -3, 0, 100\} = \max \{ \max\{10, -3\}, \max\{0, 100\} \} = 100$

- Property holds recursively (for $\max\{\mathbf{D}_1\}$ and $\max\{\mathbf{D}_2\}$)

- Why can this property be useful for Complete-Linkage ?

- Given the distances between a cluster **A** and two clusters **B** and **C** that have been merged

- It is trivial to compute the distance between **A** and $(\mathbf{B} \cup \mathbf{C})$ from the previous distances

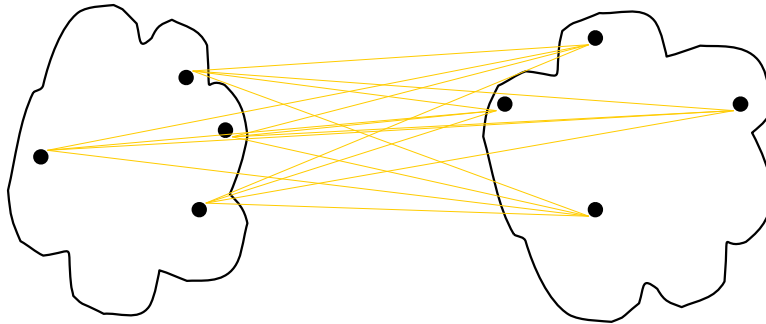
Complete-Linkage (Exercise):

- Initial distance matrix ($\mathbf{D}_1$) for 5 observations {1, 2, 3, 4, 5}

$$\mathbf{D}_1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & & & & \\ & 2 & & & \\ & 6 & 5 & 0 & \\ & 10 & 9 & 4 & 0 \\ & 9 & 8 & 5 & 3 & 0 \end{bmatrix} \end{matrix}$$

- First merge takes place between singletons 1 and 2 (closest clusters)
 - Rest is the same as for Single-Linkage, but now the distances are updated in a different way
- Exercise:** compute the **complete-linkage clustering** for these data and draw the resulting **dendrogram** (make sure you include the height at which the cluster mergers occur, which now correspond to the complete-linkage cluster distances)

How to Define Inter-Cluster (Dis)Similarity



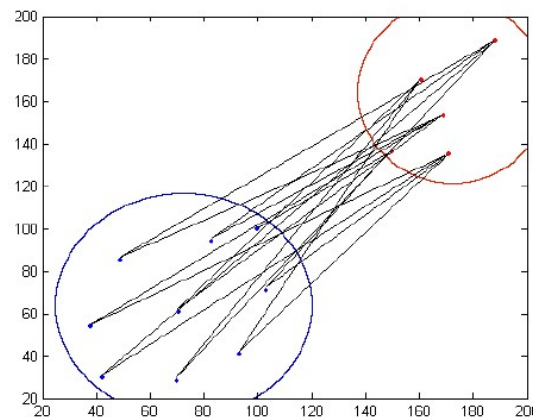
- ☐ MIN
- ☐ MAX
- ☐ **Average**
- ☐ Distance Between Centroids
- ☐ Other methods
 - Ward's
 - ...

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						

·
·
· **Proximity Matrix**

Average Linkage (UPGMA)

- Distance between *clusters* is given by the average distance between pairs of objects across the clusters in question
- It is also known as Group Average or **UPGMA**:
 - Unweighted Pair Group Method using Arithmetic averages*
 - “unweighted” → every pair of objects has the same importance



Ricardo Campello

Efficient Proximity Matrix Updates

- The (dis)similarity between a newly born cluster (formed by merging two existing clusters) and other clusters can be updated from the previously computed distances
 - rather than computed from scratch
- Specifically, let $|\mathbf{C}_i|$ be the number of objects in a cluster $\mathbf{C}_i$ and $d(\mathbf{C}_i, \mathbf{C}_j)$ be the (dis)similarity between two clusters $\mathbf{C}_i$ and $\mathbf{C}_j$. One can show that:

$$d(\mathbf{C}_i, \mathbf{C}_j \cup \mathbf{C}_k) = \frac{|\mathbf{C}_j|}{|\mathbf{C}_j| + |\mathbf{C}_k|} d(\mathbf{C}_i, \mathbf{C}_j) + \frac{|\mathbf{C}_k|}{|\mathbf{C}_j| + |\mathbf{C}_k|} d(\mathbf{C}_i, \mathbf{C}_k)$$

Exercise:

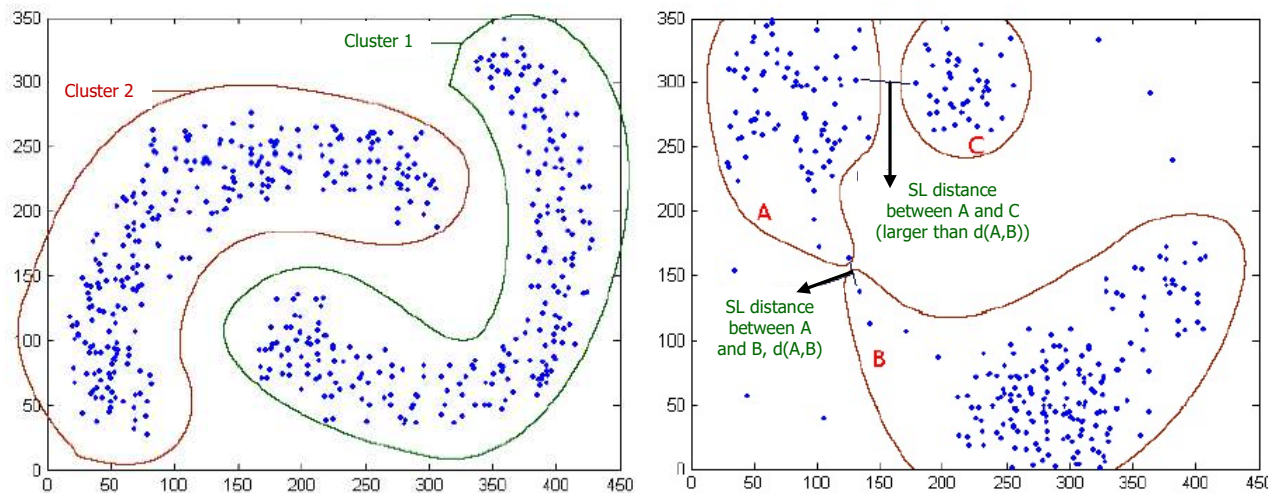
- Produce the complete dendrogram resulting from the application of the average linkage (**UPGMA**) method to the distance matrix below
- Show, step-by-step, the updated distance matrix (using the formula in the previous slide)

$$\mathbf{D}_1 = \begin{matrix} & \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} \end{matrix} \begin{bmatrix} 0 & & & & \\ 2 & 0 & & & \\ 6 & 5 & 0 & & \\ 10 & 9 & 4 & 0 & \\ 9 & 8 & 5 & 3 & 0 \end{bmatrix}$$

Summary of Single-, Complete-, and Average-Linkage

- **Single-Linkage:**

- It is capable of detecting clusters with arbitrary shapes (left figure)
- However, it is not robust to noisy data (figure on the right)



Summary of Single-, Complete-, and Average-Linkage

- Complete Linkage:

- Less sensitive to noisy data, but
 - It can be sensitive to outliers
 - It is more prone to split large clusters (even if they are "legitimate" clusters)
 - It cannot detect stretched clusters with arbitrary shapes, close to each other

- Average Linkage:

- It tends to be less sensitive to noisy data and outliers than the previous methods
- But like complete-linkage, it tends to favour "globular-shaped" clusters
- It is usually a good compromise in practice
 - Alongside with more sophisticated methods, such as Ward's and density-based algorithms (e.g. HDBSCAN*)

Ward's Method (1963)

- This method is based on the successive minimisation of the Sum of Squared Errors – SSE (cluster within variances) at each new hierarchical level (partition) as built bottom-up:

$$J = \sum_{i=1}^k \sum_{\mathbf{x}_j \in \mathbf{C}_i} d(\mathbf{x}_j, \bar{\mathbf{x}}_i)^2$$

where d = Euclidian distance and $\bar{\mathbf{x}}_i$ is the i -th cluster centre:

$$\bar{\mathbf{x}}_i = \frac{1}{|\mathbf{C}_i|} \sum_{\mathbf{x}_i \in \mathbf{C}_i} \mathbf{x}_i$$

Ward's Method

- “Dissimilarity” between pairs of clusters C_i and C_j :
 - is defined as the variation that would be observed in the SSE (J criterion) of the current partition should these clusters be merged to form the next partition
 - thus, merging the two most similar clusters is equivalent to minimising the increase in within-cluster variances at each new hierarchical level

Ward's Method

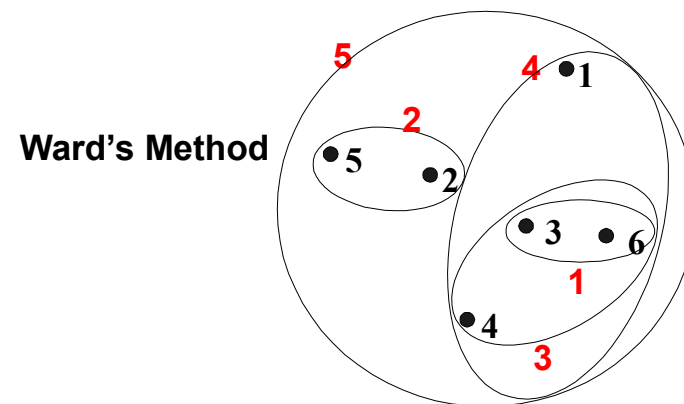
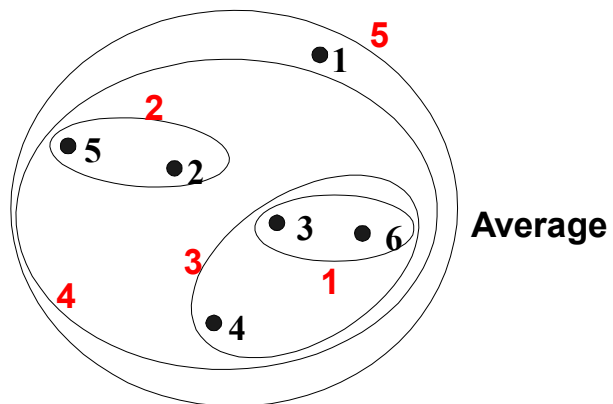
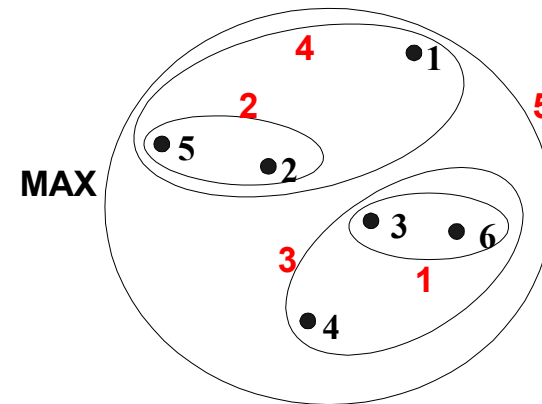
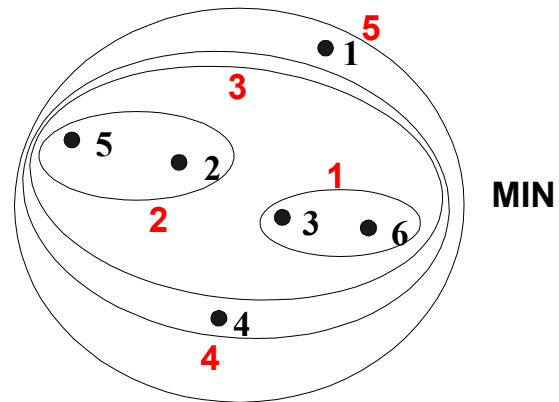
■ Cons:

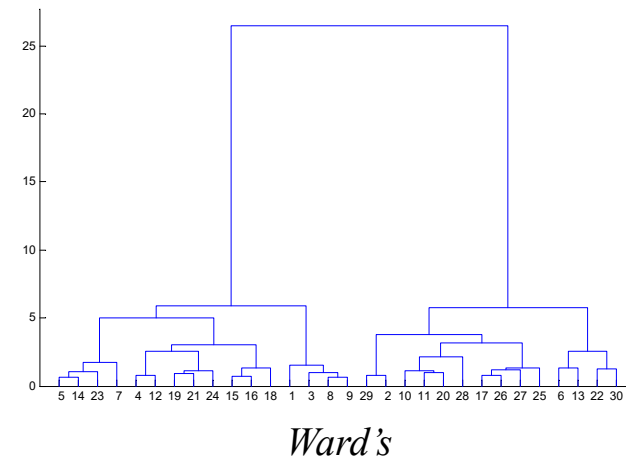
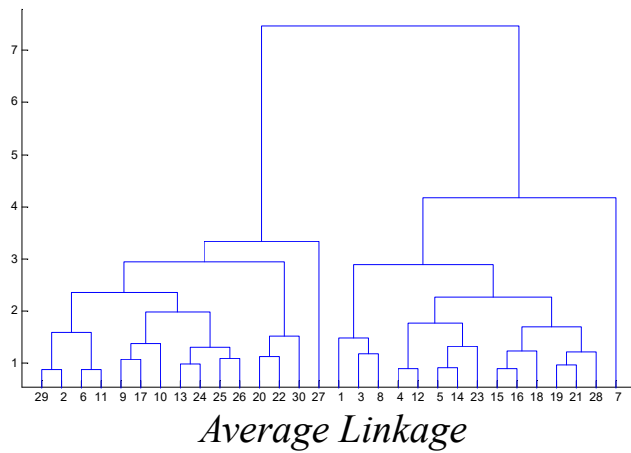
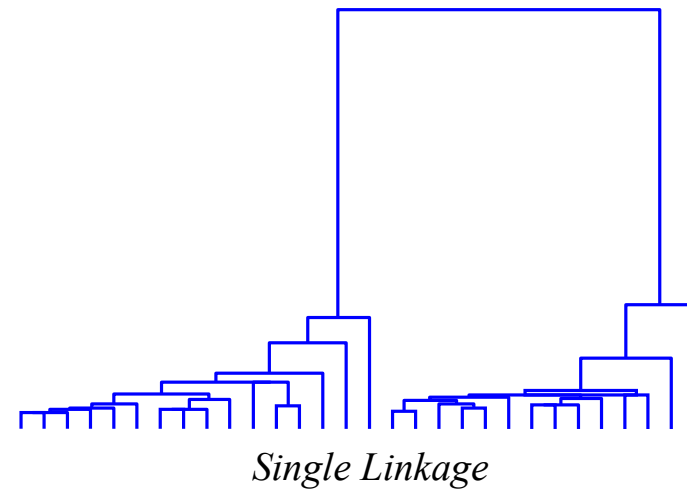
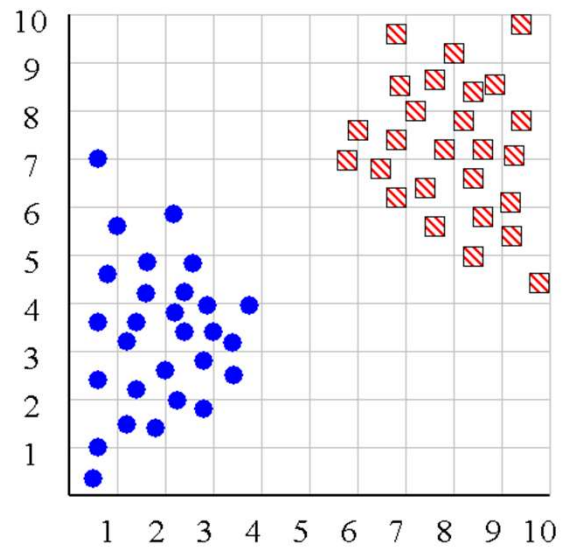
- Like Average Linkage, it tends to produce globular clusters
- Update formula is only interpretable for real-valued vector data and Euclidean distance

■ Pros:

- Similar to Average Linkage regarding robustness to noise/outliers
- "Hierarchical Counterpart" of k-means (same objective function)
 - It can be used to initialise k-means
- **Jain & Dubes (1988):** *"Several of the comparative studies discussed in Section 3.5.2 conclude that Ward's method, also called the minimum variance method, outperforms other hierarchical clustering methods"*

Hierarchical Clustering: Comparison





Keogh, E. A Gentle Introduction to Machine Learning and Data Mining for the Database Community, SBBD 2003, Brazil.

Hierarchical Clustering: Typical Time and Space Requirements

- ❑ $O(N^2)$ **memory** since it uses the proximity matrix
 - N is the number of points
- ❑ $O(N^3)$ **runtime** in many cases
 - There are N steps and, at each step, the proximity matrix must be updated and searched
- ❑ Complexity can be reduced for some approaches, e.g., Single-Linkage and Complete-Linkage

Summary of Hierarchical Methods

- **No. of Clusters**: we don't need to specify the number of clusters *in advance*, but we may need to do it *a posteriori* ...
- **Greedy Procedure**: it is not possible to fix a wrong merger in previous steps – optimal solution (e.g. in Ward's sense) is not guaranteed
- **Escalability**: running time complexity is $\Omega(N^2)$; N = no. of objects
- **Interpretability**: an entire hierarchy (rather than a partition) is produced, which is the primary goal in many applications (e.g. taxonomy) and allows data visualisation, exploratory data analysis, etc.
- **Relational Computation**: Original data is not required, since the algorithms can operate just using a distance matrix

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- ▶ Jain, A. K. and Dubes, R. C., Algorithms for Clustering Data, Prentice Hall, 1988
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- ▶ Gordon, A. D., “Hierarchical Classification”, Em Arabie et al. (Eds.), Clustering and Classification, World Scientific, 1996